

RESPIRATION IN PLANTS

90 MINUTES
SPECIALISED PYQ

Q1. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R): (NEET - 2023)

Assertion (A):	ATP is used at two steps in glycolysis .
Reason (R):	First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1,6-bisphosphate.

In the light of the above statements, choose the correct answer from the options given below :

- | | |
|----|--|
| 1. | (A) is False but (R) is True. |
| 2. | Both (A) and (R) are True and (R) is the correct explanation of (A). |
| 3. | Both (A) and (R) are True but (R) is NOT the correct explanation of (A). |
| 4. | (A) is True but (R) is False. |

Q2.Match List I with List II

(NEET - 2023)

	List I		List II
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the options given below:

Options :	A	B	C	D
1.	II	IV	III	I

(c)	ETS complex-III	(iii)	NADH dehydrogenase
(d)	ETS complex-IV	(iv)	Ubiquinone and FADH ₂ dehydrogenase

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(ii)	(i)	(iv)	(iii)
2.	(iv)	(iii)	(ii)	(i)
3.	(iii)	(ii)	(i)	(iv)
4.	(iii)	(iv)	(i)	(ii)

Q8. Identify the cytochrome which acts as a mobile carrier for the transfer of electrons between complex III and IV?
(NEET - 2022)

1.	Cytochrome a	2.	Cytochrome a ₃
3.	Cytochrome b c ₁	4.	Cytochrome c

Q9. The number of time(s) decarboxylation of isocitrate occurs during single TCA cycle is :

(NEET - 2022)

1. One
2. Two
3. Three
4. Four

Q10. The 5-C compound formed during TCA cycle is :

(NEET - 2022)

1. α -ketoglutaric acid
2. Oxalo succinic acid
3. Succinic acid
4. Fumaric acid



Q11. Which of the following statements is incorrect?

(NEET - 2021)

1.	ATP is synthesized through complex V.
2.	Oxidation-reduction reactions produce a proton gradient in respiration.
3.	During aerobic respiration, the role of oxygen is limited to the terminal stage.
4.	In ETC (Electron Transport Chain), one molecule of $\text{NADH} + \text{H}^+$ gives rise to 2 ATP molecules, and one FADH_2 gives rise to 3 ATP molecules.

Q12. The number of substrate level of phosphorylations in one turn of citric acid cycle is:

(NEET - 2020)

1.	One	2.	Two
3.	Three	4.	Four

Q13. Pyruvate dehydrogenase activity during aerobic respiration requires: (NEET - 2020)

1. Calcium
2. Iron
3. Cobalt
4. Magnesium

Q14. Conversion of glucose to glucose-6-phosphate, the first irreversible reaction of glycolysis, is catalyzed by: (NEET - 2019)

1. Phosphofructokinase
2. Aldolase
3. Hexokinase



4. Enolase

Q15. Respiratory Quotient (RQ) value of tripalmitin is:

(NEET - 2019)

1.	0.09	2.	0.9
3.	0.7	4.	0.007

Q16. Which of the following statements regarding mitochondria is incorrect? (NEET - 2019)

1.	Mitochondrial matrix contains single circular DNA molecule and ribosomes.
2.	Outer membrane is permeable to monomers of carbohydrates, fats, and proteins.
3.	Enzymes of electron transport are embedded in the outer membrane.
4.	Inner membrane is convoluted with infoldings.

Q17. Where is the respiratory electron transport system (ETS) located in plants? (NEET - 2019)

1. Mitochondrial matrix
2. Outer mitochondrial membrane
3. Inner mitochondrial membrane
4. Intermembrane space

Q18. What is the role of NAD⁺ in cellular respiration?

(NEET - 2018)

1.	It functions as an enzyme
2.	It functions as an electron carrier
3.	It is a nucleotide source for ATP synthesis
4.	It is the final electron acceptor for anaerobic respiration

Q19. Which of these statements is incorrect?

(NEET - 2018)

1.	Enzymes of the TCA cycle are present in the mitochondrial matrix.
2.	Glycolysis occurs in the cytosol.
3.	Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.
4.	Oxidative phosphorylation takes place in the outer mitochondrial membrane.

Q20. Which statement is wrong for Krebs' cycle?

(NEET - 2017)

1.	There is one point in the cycle where FAD^+ is reduced to FADH_2 .
2.	During conversion of succinyl CoA to succinic acid, a molecule of GTP is synthesized.
3.	The cycle starts with condensation of acetyl group (acetyl CoA) with pyruvic acid to yield citric acid.
4.	There are three points in the cycle where NAD^+ is reduced to $\text{NADH} + \text{H}^+$.

Q21. Which of the following biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins?

(NEET - 2016)

1.	Glucose-6-phosphate
2.	Fructose 1, 6-bisphosphate
3.	Pyruvic acid
4.	Acetyl Co-A



Q22. Oxidative phosphorylation is

(NEET - 2016)

1.	formation of ATP by transfer of phosphate group from a substrate to ADP
2.	oxidation of phosphate group in ATP
3.	addition of phosphate group to ATP
4.	formation of ATP by energy released from electrons removed during substrate oxidation

Q23. In which one of the following processes CO₂ is not released?

(AIPMT - 2014)

1. Aerobic respiration in plants
2. Aerobic respiration in animals
3. Alcoholic fermentation
4. Lactate fermentation

Q24. Which of the metabolites is common to respiration mediated breakdown of fats, carbohydrates and proteins?

(AIPMT - 2013)

1. Fructose 1, 6 - biphosphate
2. Pyruvic acid
3. Acetyl CoA
4. Glucose- 6 -phosphate

Q25. In mitochondria, protons accumulate in the:

(AIPMT - 2011)

1. Matrix
2. Outer membrane
3. Inner membrane
4. Intermembrane space



(AIPMT - 2010)

- (AIPMT - 2009)

- (AIPMT - 2007)

(AIPMT - 2007)

- (AIPMT - 2006)

Q31. Emerson's enhancement effect and red drop have been instrumental in the discovery of?

ANSWER KEY

1. (2)	11. (4)	21. (4)
2. (1)	12. (1)	22. (4)
3. (4)	13. (4)	23. (4)
4. (1)	14. (3)	24. (3)
5. (1)	15. (3)	25. (4)
6. (4)	16. (3)	26. (2)
7. (4)	17. (3)	27. (3)
8. (4)	18. (2)	28. (4)
9. (2)	19. (4)	29. (1)
10. (1)	20. (3)	30. (4)

